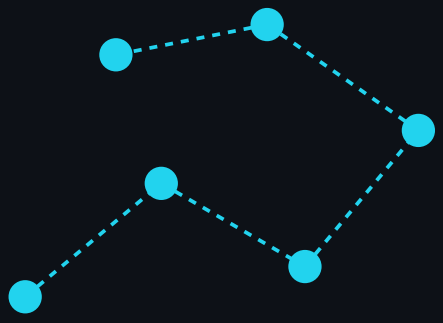


Gephyra

Graphical interface - user manual

Water-bridge network discovery, clustering and visualization

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https://github.com/insilichem/Gephyra

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1. What the interface does

Gephyra finds chains of water molecules that connect a chosen *root* atom (for example a ligand oxygen) to the surrounding solvent in a molecular dynamics trajectory, groups the chains that recur into spatial channels, and writes ready-to-run scenes for molecular viewers.

The graphical interface runs the whole workflow in three steps, without memorising command-line options:

- **1. Pathway Analysis** reads the topology and trajectory and writes every water bridge, frame by frame, to a results file (`.jsonl`).
- **2. Clustering** groups the bridges that recur in space into channels and writes a cluster file (`.json`).
- **3. Visualization** writes a script for UCSF ChimeraX, UCSF Chimera, PyMOL or VMD that loads the protein, the ligand and the water network and shows them together.

Every step shows its progress live, can be cancelled, and ends with a confirmation that lists the files written and the folder they are in.

2. Installation and start-up

Requirements

- **Python 3.10 or newer; 3.11 recommended**, in a dedicated conda environment (below). Using a separate environment keeps the Python version and libraries Gephyra needs from clashing with other software, such as a PyMOL installation that ships its own Python.
- Conda (Miniconda, Miniforge or Anaconda) to create that environment.
- MDAnalysis, NetworkX, NumPy and SciPy, installed together with Gephyra.
- At least one molecular viewer to look at the results: UCSF ChimeraX, UCSF Chimera, PyMOL or VMD.

Installing Gephyra

In a terminal, download Gephyra, create its environment and install it:

```
git clone https://github.com/insilichem/Gephyra.git
cd Gephyra
conda create -n gephyra -c conda-forge python=3.11 "tk=*=xft*"
conda activate gephyra
pip install .
```

The same environment, with the scientific libraries also taken from conda-forge, can be created from the file shipped with Gephyra:

```
git clone https://github.com/insilichem/Gephyra.git
cd Gephyra
conda env create -f environment.yml
conda activate gephyra
pip install .
```

`"tk=*=xft*"` selects the build of the Tk graphics library with smooth, anti-aliased fonts; with the default build the interface text looks blocky. Check the installation with `gephyra --version`.

Tip. Run `conda activate gephyra` in every new terminal before using Gephyra. If the `gephyra` command is not found, the environment is not active.

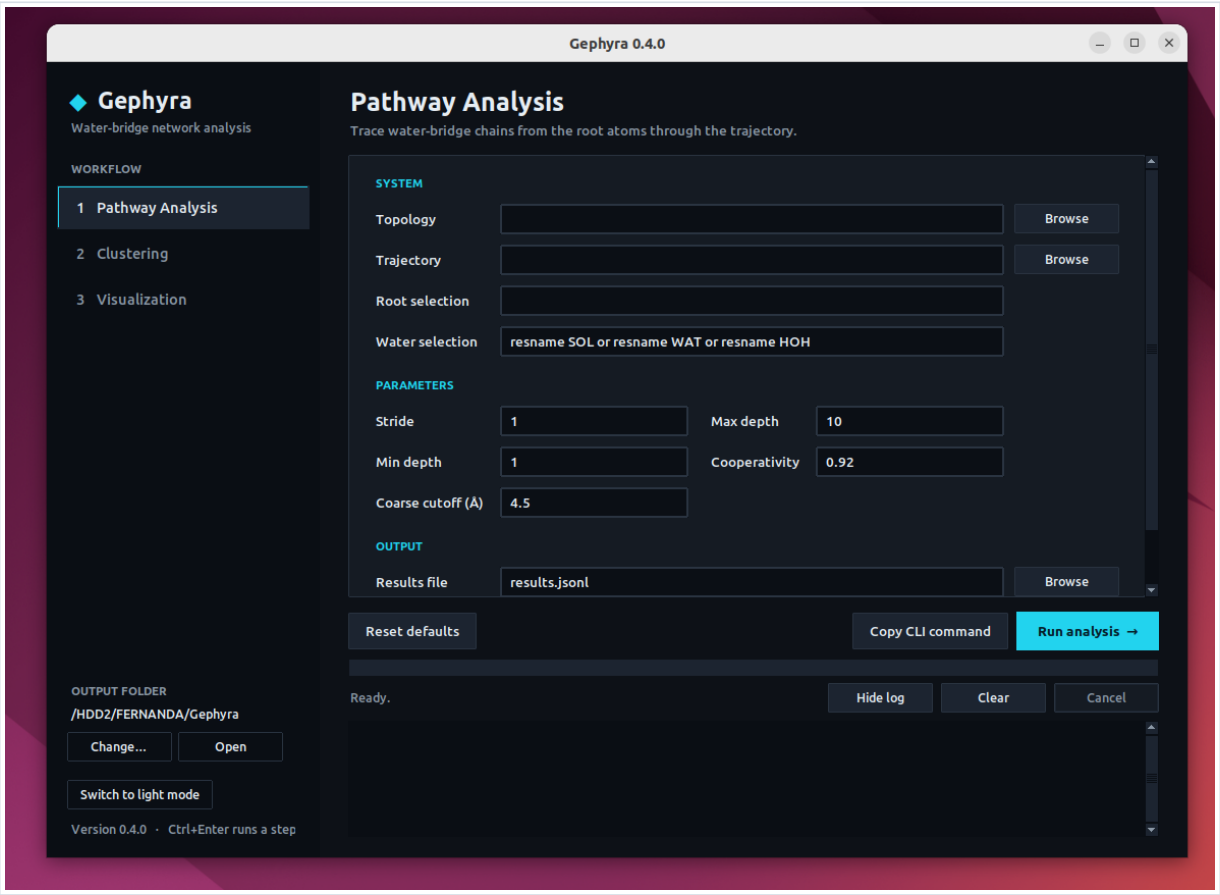
Starting the interface

```
conda activate gephyra
gephyra gui
# or
gephyra-gui
```

The window opens on step 1. The folder you start it from is the default output folder (you can change it at any time, see section 3).

Tip. If the text looks blocky or pixelated, Gephyra is running in a Python whose Tk library has no anti-aliased fonts, and the interface says so in its log. This does not happen in the environment above; in another conda environment, install the right build with `conda install -c conda-forge "tk=*=xft*"` and restart the interface.

3. The window at a glance

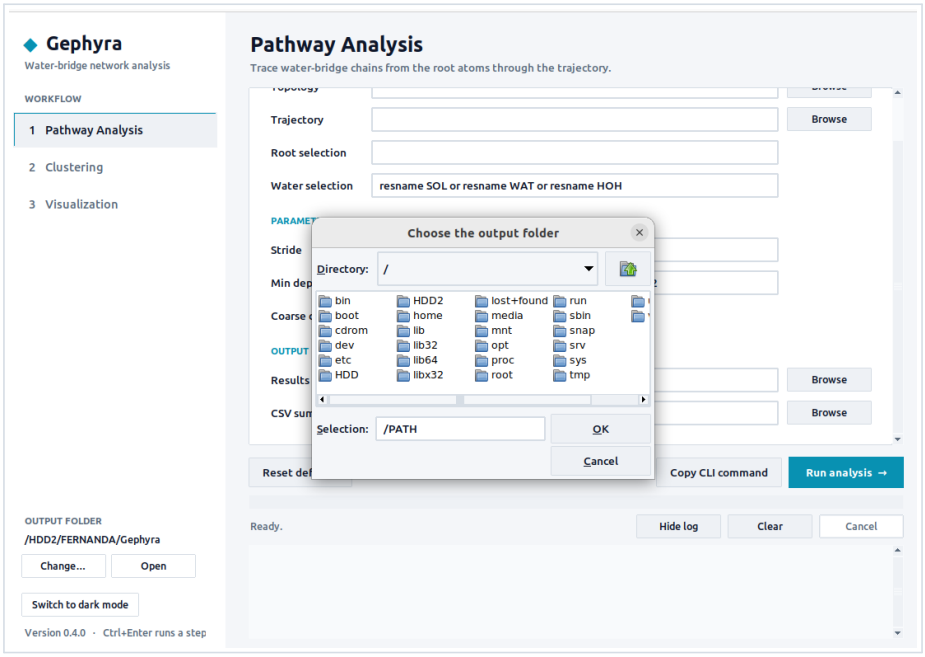


The main window: sidebar with the three steps and the output folder, the form, the progress bar and the log.

Area	What it does
Sidebar: workflow	The three steps. Click one to open its form; the current step is highlighted.
Sidebar: output folder	Where all results are saved. **Change...** picks another folder, **Open** shows it in your file manager.
Sidebar: theme	Switches between the dark and the light theme. The choice is remembered.
Form	The options of the current step, grouped in sections. Hover over a field to see a short explanation. **Browse** opens a file picker.
Buttons	**Reset defaults** restores the standard values, **Copy CLI command** copies the equivalent command line, and the highlighted button runs the step (also Ctrl+Enter).
Progress bar and status	Show what is running and how far it has got. **Cancel** stops a running step after the current frame.
Log	Messages from Gephyra as they happen: warnings in amber, errors in red, saved files in green. **Hide log** and **Clear** manage it.

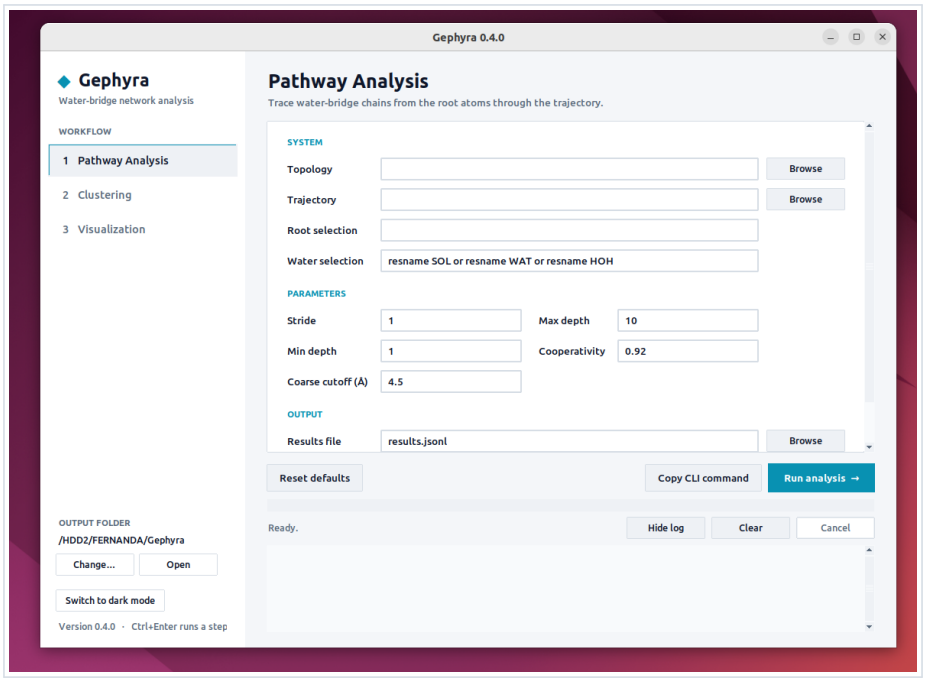
Where results are saved

Output fields such as *Results file* or *Script name* hold file **names**, which are saved inside the output folder shown in the sidebar. Type an absolute path in a field to save that file somewhere else. If the output folder does not exist, the interface offers to create it.



Choosing the output folder with Change... in the sidebar.

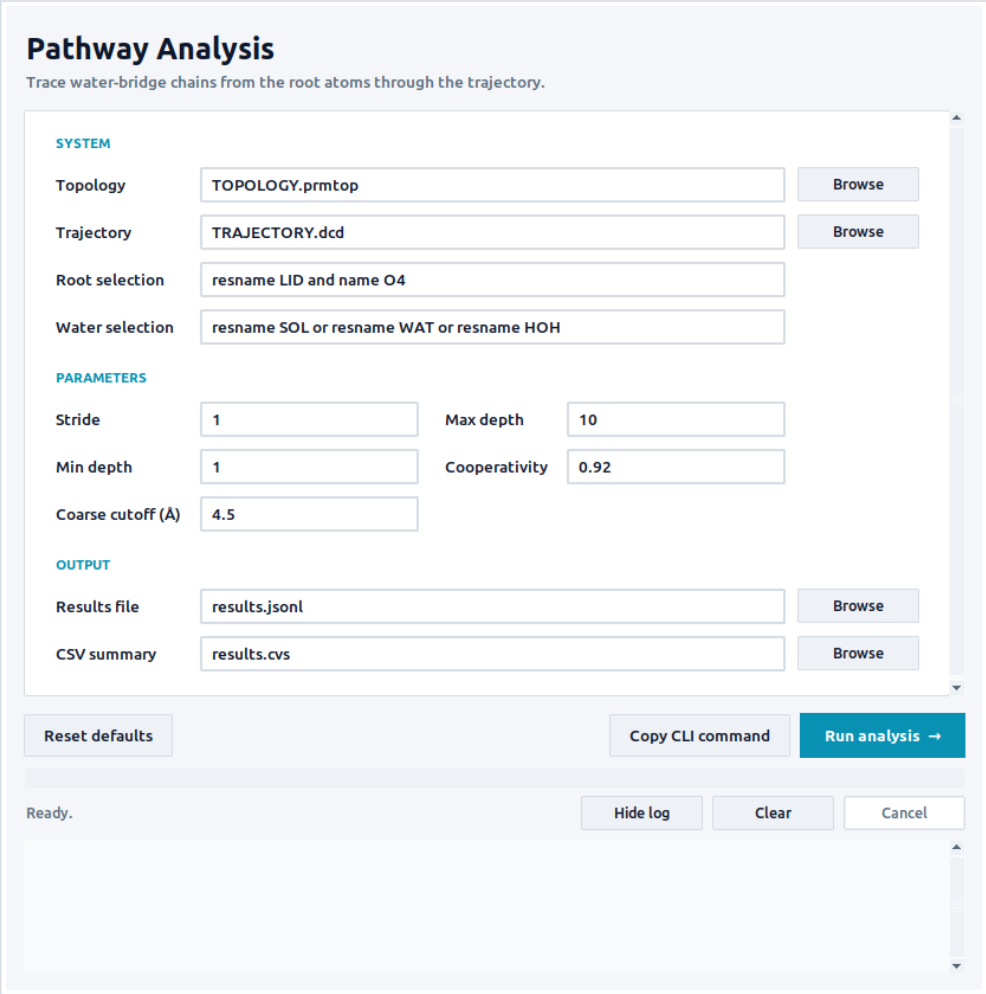
After a step finishes, its confirmation window lists every saved file with its folder, and offers **Open folder** and **Save a copy to...**, so results can also be stored in another location after the calculation.



The same window in the light theme.

4. Step 1 - Pathway Analysis

Finds the water bridges in every analysed frame of the trajectory.

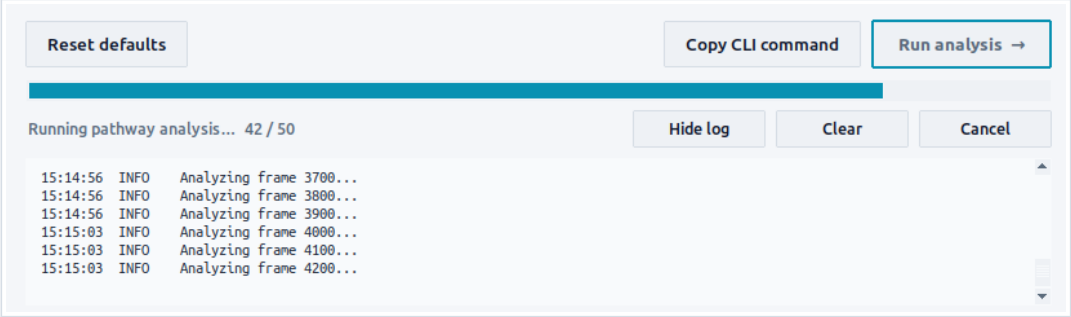


Step 1, Pathway Analysis, filled in with a topology, a trajectory and a root selection.

Field	Meaning	Default
Topology	Structure/topology file of the system (.prmtop, .psf, .tpr, .pdb, ...). Bond information enables the full hydrogen-bond geometry criterion.	required
Trajectory	The frames to analyse (.dcd, .xtc, .nc, ...). For a single structure, choose a coordinate file such as a .pdb.	required
Root selection	MDAnalysis selection of the atom(s) the bridges start from, e.g. `resname LIG and name O4`.	required
Water selection	MDAnalysis selection of the bridging solvent.	SOL, WAT or HOH
Stride	Analyse every N-th frame. Larger values are faster.	1
Max depth / Min depth	Longest and shortest chains (number of waters) that are kept.	10 / 1
Cooperativity	Per-step discount along a chain, between 0 and 1 (1 switches it off).	0.92
Coarse cutoff	Neighbour search radius in angstrom.	4.5

Field	Meaning	Default
Superpose on	Atoms used to superpose every frame on the first one, so that the same water channel has the same coordinates in every frame even if the protein drifts or rotates in the box. *auto* uses the protein C-alpha atoms; *none* keeps the trajectory coordinates (only for trajectories that are already fitted).	auto
Results file	Name of the frame-by-frame results file in the output folder.	results.jsonl
CSV summary	Optional table of all paths, for spreadsheets.	-

Tip. For a first look at a long trajectory, start with a stride of 10-50. The analysis time grows with the number of frames and with Max depth.



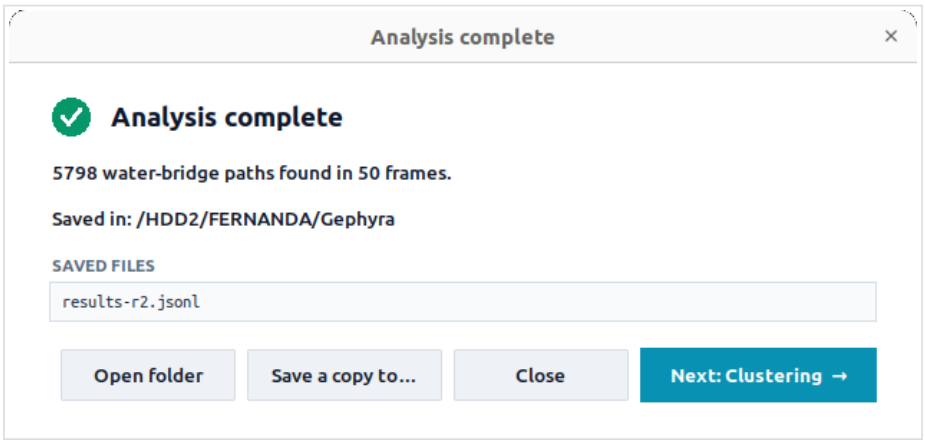
While the analysis runs: the progress bar counts frames and the log streams messages.

Runtime estimate

The analysis is the slowest step. When you click **Run analysis**, Gephyra first times a few sample frames of your system (this takes a few seconds) and shows how long the whole run will take, together with the size of the system: atoms, residues, water molecules, trajectory frames and how many of them the stride selects. If the run would be long it also suggests how to shorten it, for example the stride that would bring it to about ten minutes, or a smaller Max depth. Choose **Run analysis** to start or **Cancel** to change the settings first.

Tip. What costs time is mostly the hydrogen-bond network around the root atom, which grows with Max depth (each extra water layer adds a new shell of waters to score) and with the number of analysed frames. System size matters less. While the analysis runs, the status line shows how much time is left.

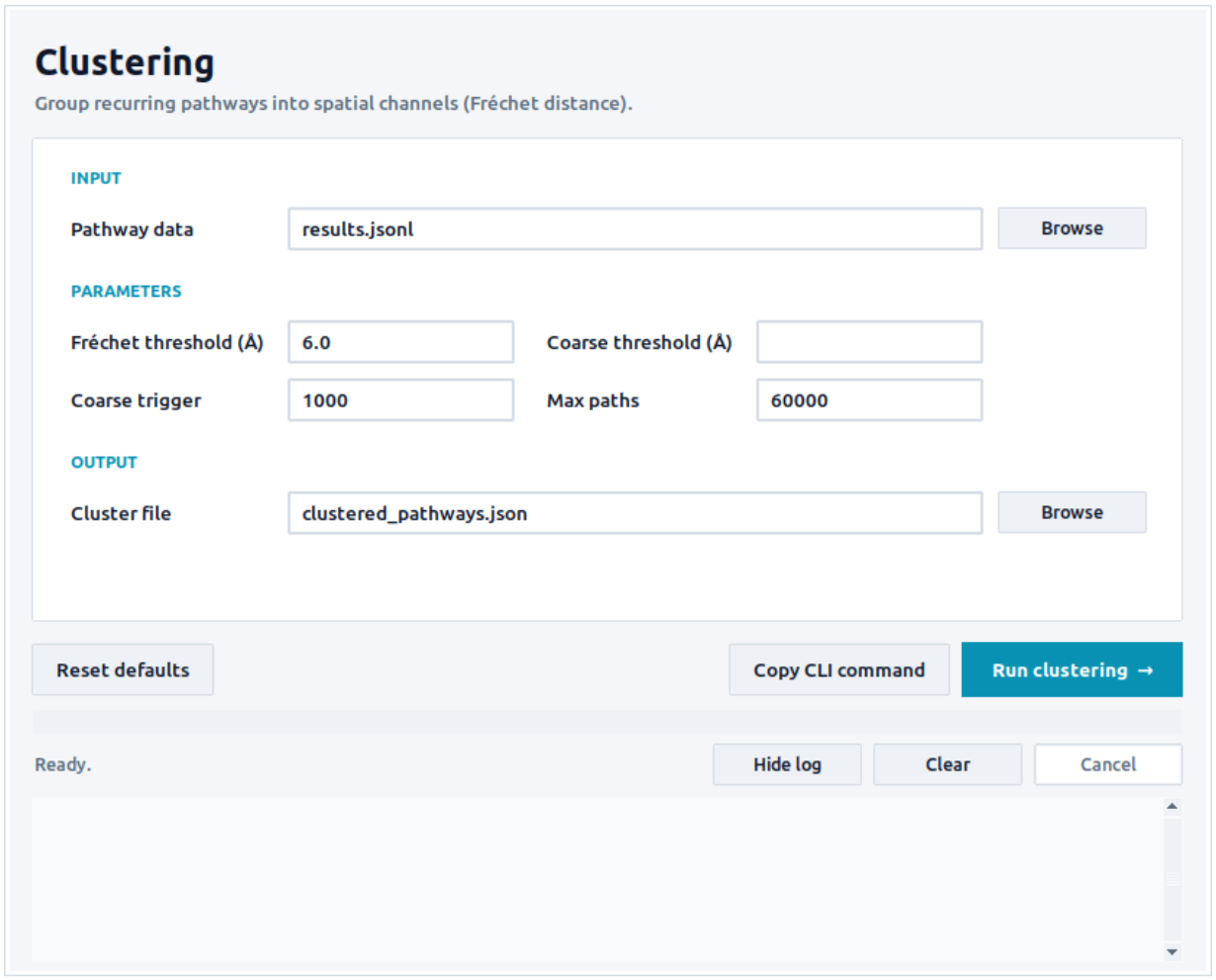
Click **Run analysis**. The progress bar counts the frames; when it finishes, a confirmation reports how many water-bridge paths were found and where the files were saved. Click **Next: Clustering** to continue: the results file is already filled in on the next step.



The confirmation shown when the analysis finishes, listing the saved files and their folder.

5. Step 2 - Clustering

Groups the paths found in step 1 into recurring spatial channels. Different water molecules occupying the same channel at different times end up in the same cluster.



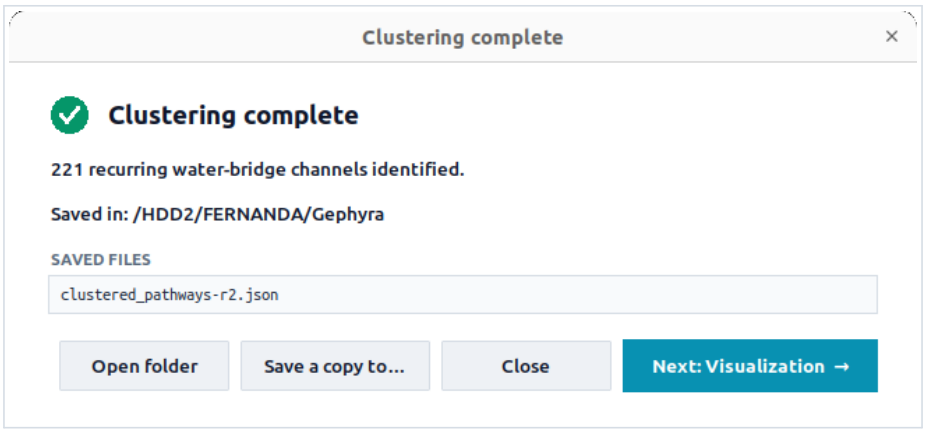
Step 2, Clustering, with the pathway data filled in from step 1.

Field	Meaning	Default
Pathway data	The results file of step 1 (filled in automatically).	-
Frechet threshold	How similar two paths must be (in angstrom) to belong to the same cluster. Lower values give more, finer clusters; 3-8 is a typical range.	6.0
Coarse threshold / trigger	Settings of a fast pre-screening used for very large data sets; the defaults suit most cases.	auto / 1000
Max paths	Memory cap: beyond this, only the most frequent paths are clustered.	60000
Cluster file	Name of the cluster file in the output folder.	clustered_pathways.json



While clustering runs: progress through the Frechet distance matrix.

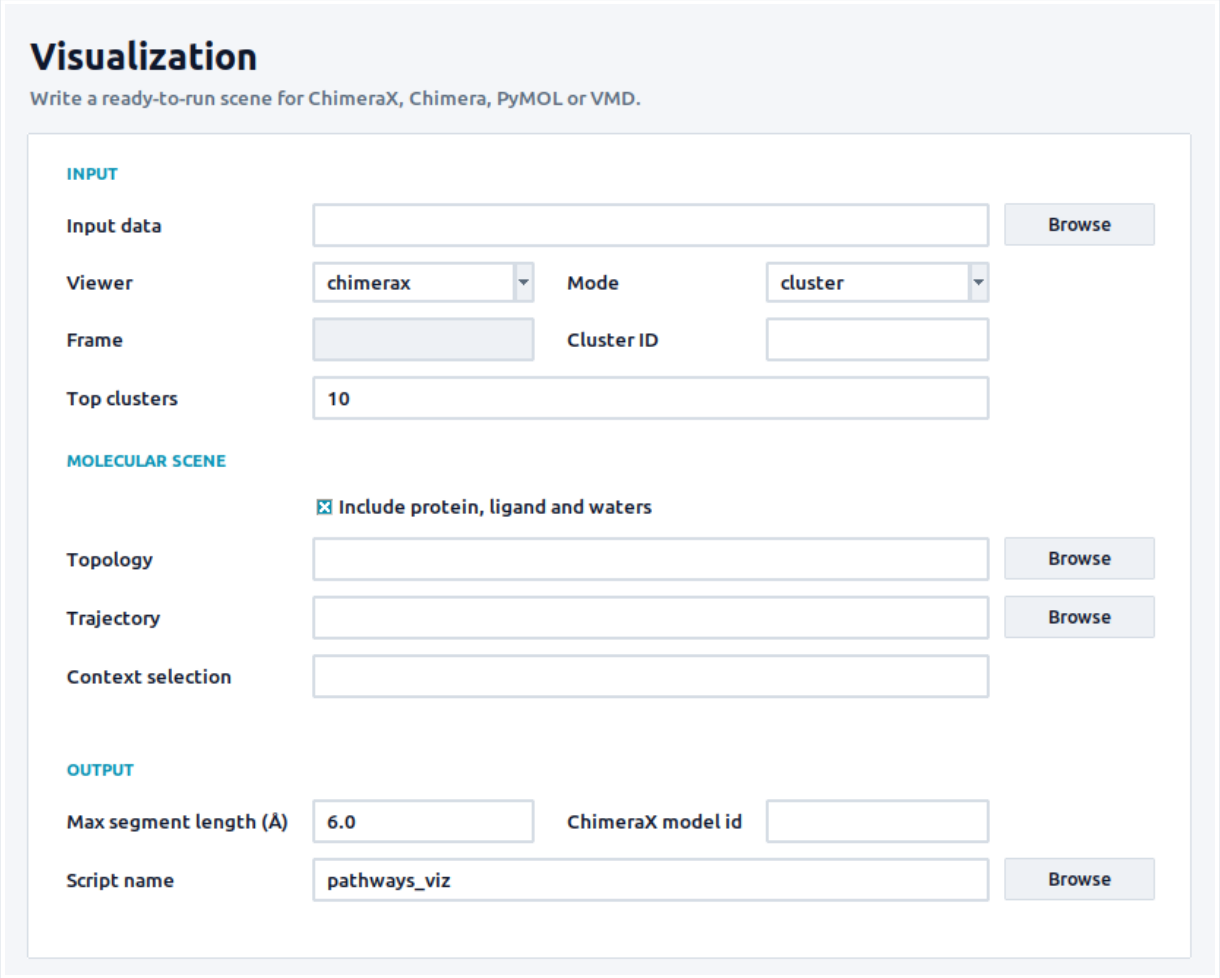
Click **Run clustering**. When it finishes, a confirmation window reports how many channels were identified and lists the saved file. **Next: Visualization** opens step 3 with the cluster file already selected.



The confirmation shown when clustering finishes.

6. Step 3 - Visualization

Writes a script that opens a complete scene in your molecular viewer: the protein as a cartoon, the ligand and the bridging waters as sticks, any cofactor or metal complex as sticks (metal ions as spheres), and the water network drawn on top.



Step 3, Visualization: viewer, mode and the molecular-scene section.

Choosing what to draw

Mode	Input file	What the scene shows
cluster	cluster file (step 2)	The protein and ligand with the medoid (most representative) path of each cluster as a colour-coded tube, and a legend with each cluster's occupancy. **Top clusters** limits the picture to the most occupied ones. Enter a **Cluster ID** to rebuild that one cluster as real water molecules at the frame it comes from.
frame	results file (step 1)	One frame: the real water molecules (oxygen and hydrogens) and every water bridge as a dashed bond. Choose the frame number in **Frame**; leave it empty to draw the first frame with water bridges. If a frame has none, the message lists the frames that do.
density	results file (step 1)	Every path of every frame as thin transparent tubes, showing where water bridges form most often.

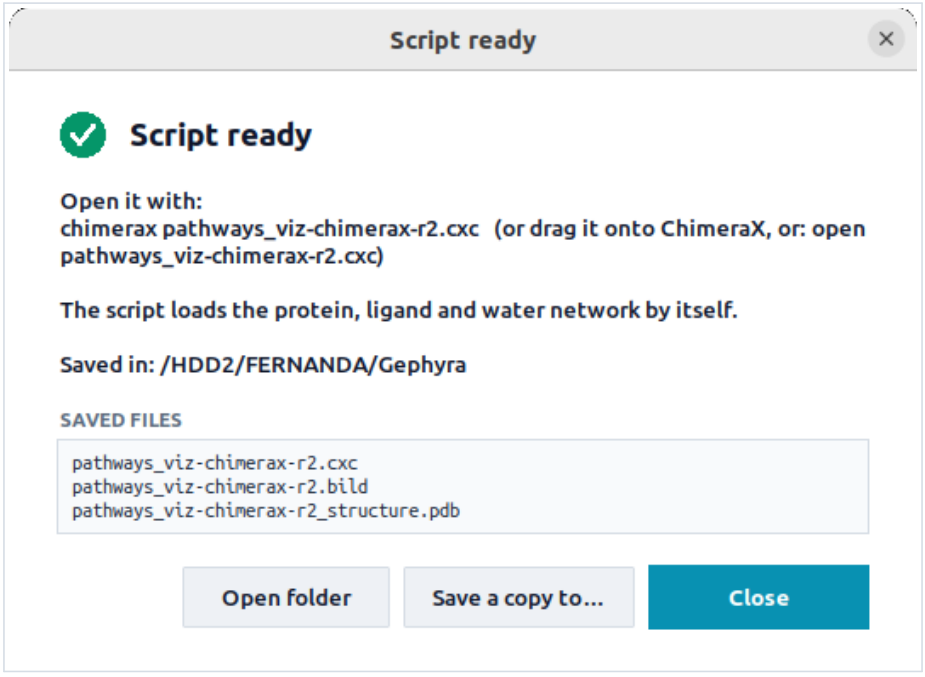
Molecular scene

With **Include protein, ligand and waters** ticked (the default), Gephyra re-opens the analysed trajectory and writes a small structure file (`<name>_structure.pdb`) next to the script, which the script loads by itself. The topology and trajectory are read from the data file; the line under the section says which ones. Fill in **Topology** and **Trajectory** only for older result files that do not record them, or if the files have moved.

Untick it to write only the network geometry, then open the analysed structure in the viewer yourself before running the script.

Writing and opening the script

Pick the **Viewer**, click **Write script**, and the confirmation shows the exact command that opens it. Do not open the structure first: the script does it.

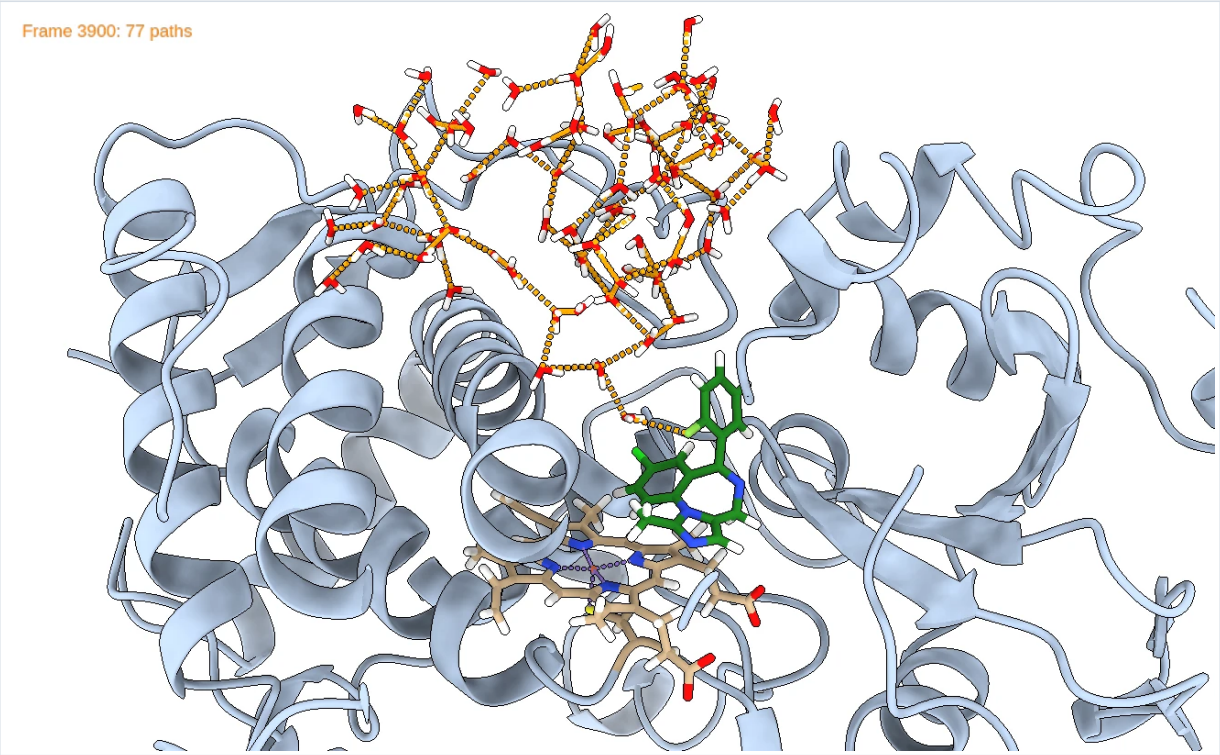


The 'Script ready' confirmation with the command that opens the script.

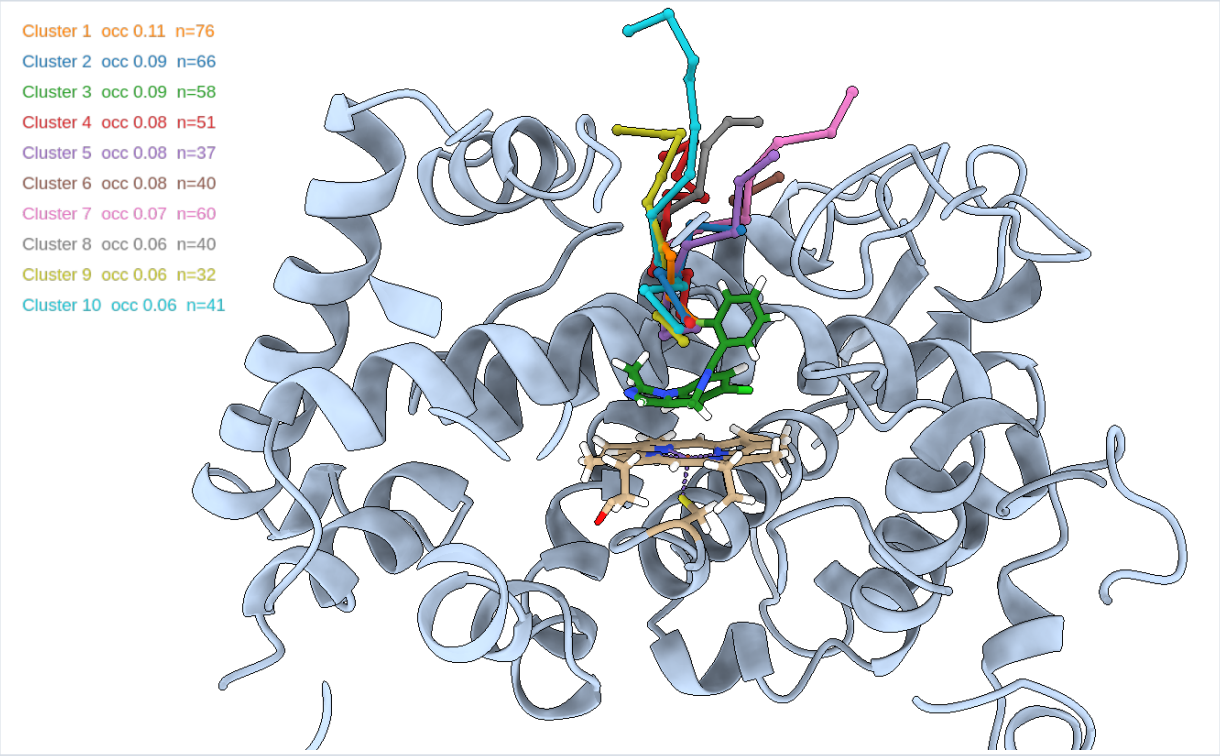
Viewer	Script	How to open it
UCSF ChimeraX	.cxc	`chimerax scene.cxc`, drag the file onto the window, or type `open scene.cxc`.
UCSF Chimera	.py	`chimera scene.py`, or File > Open...
PyMOL	.py	`pymol scene.py`, or `run scene.py` inside PyMOL.
VMD	.tcl	`vmd -e scene.tcl`, or `source scene.tcl` in the Tk Console. The `-e` matters: without it VMD tries to read the script as a molecule.

VMD scripts must be run with `vmd -e scene.tcl`. Without `-e`, VMD treats the script as a molecule file, reports *Unable to determine file type*, and draws nothing. Inside a running VMD, use `source scene.tcl` in Extensions > Tk Console instead.

Running a script again replaces the previous Gephyra scene.



A frame scene in ChimeraX: protein cartoon, ligand, water molecules and dashed water bridges.



A cluster scene: colour-coded medoid paths around the protein, with the legend.

7. Output files

File	Written by	Contents
results.jsonl	Step 1	One line per analysed frame with every water bridge (atoms, coordinates, H-quality score), plus the analysed topology/trajectory and settings.
summary.csv	Step 1 (optional)	The same paths as a table.
clustered_pathways.json	Step 2	The channels: size, occupancy, persistence, H-quality and the medoid path of each, plus the analysed system.
name.cxc / .py / .tcl	Step 3	The viewer script.
name_structure.pdb	Step 3	The protein, ligand and bridging waters loaded by the script.
name.bild, name.pb	Step 3	Network geometry and pseudobonds used by the Chimera and ChimeraX scripts.

H-quality (Z) sums the geometric quality of every route through a channel. It is not a probability and can exceed 1 for channels with many parallel routes; *hquality_norm* = Z/(1+Z) is a bounded version for comparisons. See the README for details.

8. Troubleshooting

Message or symptom	What to do
Text looks blocky or pixelated	Install the anti-aliased Tk build: `conda install -c conda-forge "tk=*=xft*"` (section 2).
Root selection returned no atoms	Check the residue and atom names in your topology; the selection uses MDAnalysis syntax.
A trajectory file is required	Choose a trajectory. For a single structure, choose a .pdb as the trajectory.
Mode 'cluster' needs the cluster file	In step 3, cluster mode reads the file from step 2; frame and density modes read the results file from step 1.
The data file does not record its structure	The file comes from an older version of Gephyra. Fill in Topology and Trajectory in the Molecular scene section.
VMD: Unable to determine file type	Open the script with `vmd -e scene.tcl`.
Clustering finds no channels	Check that the trajectory frames are superposed: keep **Superpose on** at *auto* (or a selection of rigid atoms) and repeat step 1. Also check how many frames contain bridges at all: a channel must appear in at least two frames to be kept, so for rare bridges use a smaller stride.
The analysis takes too long	Use the runtime estimate shown before the run: a larger stride or a smaller Max depth (6 is often enough) shortens it most. From the command line, add `--estimate` to get the estimate without running.
Too many clusters to read the picture	Lower **Top clusters**, or draw a single cluster with **Cluster ID**.
Bridges or protein cut at the box edge	Image/centre the trajectory before the analysis (for example cpptraj autoimage or gmx trjconv -pbc mol -center).
Something else went wrong	Start the interface with `gephyra -v gui` to see full error details in the terminal, and include them when reporting the problem.

9. Command-line equivalents

Every run writes its command-line equivalent to the log, and **Copy CLI command** copies it, so a setup made in the interface can be repeated in scripts or on a cluster. For example:

```
gephyra calculate --topo system.prmtop --traj run.dcd \
    --root "resname LIG and name O4" --stride 20 --output results.jsonl
gephyra cluster --data results.jsonl --output clustered_pathways.json
gephyra visualize --data clustered_pathways.json --format chimerax \
    --mode cluster --top_clusters 10 --output cluster_scene
```

10. Video walkthrough

A short video walkthrough of the interface will be linked here.